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Weak sustainable development trajectories and evolving organisational physiologies: Empirical evidence from Greece

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Weak sustainable development trajectories and evolving organisational physiologies: Empirical evidence from Greece

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1. INTRODUCTION

This chapter's aim is threefold. First, we will examine the physiological transformation of firms located in a regional business ecosystem that is less robust in terms of sustainable development. Second, we will investigate whether the apparent "dominance" of monad-centric physiology —relatively instinctive strategic decisions, somewhat sporadic technological choices, and management based extensively on day-to-day practice— is related to the underperformance of regions with similar dynamics and prospects. Third, we will derive general theoretical guidelines for helping these socioeconomic systems evolve, setting the stage for considering meso–micro policies to reinforce these weak (less developed and competitive) organisations.

This research seems to fill a gap in the literature on empowering entrepreneurship and strategy-making in less-developed regions. The relevant literature tends to distinguish firms according to their size, looking mainly at the dynamic characteristics of small- and medium-sized enterprises. Thus, it overlooks, to some extent, the organic developmental problems highlighted by the organisational physiology approach (cf. Belyaeva & Lopatkova, 2020; Karagouni, 2018; Lampadarios & Kyriakidou, 2020; Migliaccio & Rossetti, 2020). Instead, this chapter synthesises different surveys we have implemented in some less developed companies,

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which we have approached from a biological perspective. Thus, it seems that this research fills a relative gap in the literature by attempting to discern the deeper evolving causes of their comparative underdevelopment —such a study seems necessary given that beneath seemingly identical quantities can lie significant differences in qualitative dimensions (e.g., Montanari & Kocollari, 2020).

The remainder of the chapter is structured as follows. Section 2 reviews relevant literature and introduces the conceptual framework. Next, Section 3 captures and analyses the methodology for approaching these research objectives. Section 4 presents a) secondary macro–meso socioeconomic data concerning a less developed Greek region (Eastern Macedonia–Thrace) and b) results of field research examining the physiology of specific organisations in this business ecosystem. The concluding Section 5 discusses and critically integrates the results of this empirical investigation. It also presents research limitations, prospects for enriching the findings, and policy directions.

2. LITERATURE REVIEW

We cannot define a caterpillar and then use the same definition for a butterfly [Edith Penrose, "The theory of the growth of the firm," a book published in 1959.]

Marshall (1890), although a forerunner of neoclassical economics, was one of the first to relate evolutionary–Darwinian biology to economics, understanding that *natura non facit saltum*. In modern evolutionary economics, biological metaphors are a *conditio sine qua non* for a dialectical–evolutionary view of the firm (Nelson et al., 2018). In the evolutionary theory of economics and political economy, the firm responds proactively to changes in the external environment (Nelson & Winter, 1982). This actor is at the heart of socioeconomic dynamics, adapting according to the skills and dynamic capabilities it articulates over time (Hodgson &

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Knudsen, 2004; Teece et al., 1997). Also, some approaches conceptualise the evolution of these entrepreneurial organisations in biological and ecosystem terms (e.g., Foster, 1997; Witt, 2006). According to Vlados (2004), firms resemble biological species with distinct organisational physiologies due to unique internal Stra.Tech.Man (Strategy–Technology–Management) compositions. The author suggests that a) organisational strategy is about where we are and why we desire a direction toward the targeted destination, b) technology is about how and why we create, diffuse, and reproduce expertise, and c) management is about how and why we manage resources in particular ways (Vlados, 2004). Evolutionary organisational physiology —in terms of Stra.Tech.Man— is path-dependent, as distinct past choices of strategy, technology, and management determine their future development (e.g., Chatzinikolaou & Vlados, 2019).

Organisational physiology captures a micro actor's evolution and sustainability prospects, meaning its ability to address current pressing problems by considering future profound transformations of the newly emerging global economy (e.g., Onyama, 2021; Vargas-Hernández & Orozco Quijano, 2022). This physiological approach leads to a typology of different entrepreneurship structures. In Vlados' (2004) view, there are three main types of entrepreneurial physiology: monad-centred, massive, and flexible. Business monad-centredness refers primarily to instinctive strategic decisions, sporadic technological choices, and management based solely on day-to-day practice. Business massiveness means strategy based mainly on mechanistic-oriented performance measurement methods, linear technology, and highly specialised, narrowly focused management. Business flexibility corresponds to an evolutionary-historical strategy, increasingly networked technology, and management constantly tending toward participatory forms. Based on this approach, the Greek entrepreneurial ecosystem appears to be home primarily to monad-centred firms, which are at the root of Greece's chronic competitiveness lags (Vlados et al., 2021).

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In this study, we empirically test some of the relevant aspects of the Stra.Tech.Man approach. Figure 1 illustrates the synthesising theoretical background we introduce based on this perspective.

<Figure 1 about here>

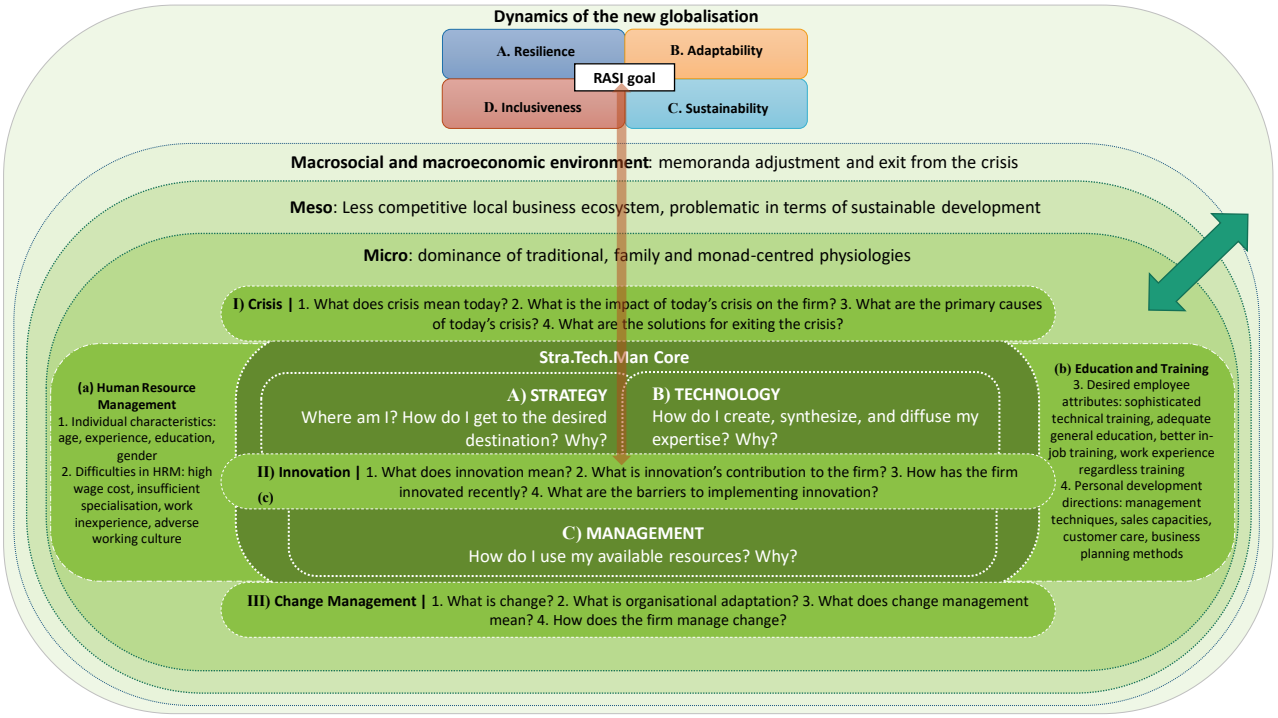


Figure 1. The conceptual framework. Source: Authors' own design.

Evolutionary economics studies socioeconomic systems as interactions of macro–meso–micro determinants (e.g., Dopfer et al., 2004). The new globalisation that emerges today unifies the macrosocial and macroeconomic levels, structurally transforming all actors and spatial systems (Chatzinikolaou & Vlados, 2022b). In particular, the new globalisation marks the critical transition in the global economy that began to unfold around 2008 —although incubated earlier through the structural maturation of the previous regime (globalisation). The financial crisis of 2008, the pandemic of 2019–2020 that hastened the fourth industrial revolution, and the Russian invasion in Ukraine of 2022 that expedited the ongoing energy transition are milestones of the emerging new globalisation (Chatzinikolaou & Vlados, 2022b). Also, since we will be

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studying a Greek region, we mention that the macroenvironment in Greece has been rapidly adjusted in recent years. Greek governments signed successive memoranda of understanding with the International Monetary Fund and relevant European institutions as loan agreements that contributed to the gradual exit from the internal structural crisis (e.g., Andrikopoulos & Nastopoulos, 2015).

Concerning today’s emerging new globalisation, the relevant literature mainly distinguishes low-performance scenarios for the near future (e.g., Bhattacharya et al., 2017; Killian, 2021). In other research we are currently conducting on innovation and change management for the new globalisation, we have evidence that the RASI synthesis (Resilience–Adaptability–Sustainability–Inclusiveness) should be the long-term goal of all actors as the international community appears to be increasingly sensitive to these issues (e.g., World Economic Forum, 2018).

Regarding the Stra.Tech.Man organisational core, we suggest this as the primary diagnostic mechanism —it is the epicentre that generates innovation in the different physiological types of firms (monad-centred, massive, or flexible). In this conceptual framework that we introduce, the achievement of RASI goals structurally reshapes global development prospects and the sustainability of local–regional business ecosystems, which include interdependent firms (Moore, 1993; Rinkinen & Harmaakorpi, 2018). The Greek economy is seemingly dominated by traditional, family-centred, and monad-centred physiologies, leading local business ecosystems to problems of competitiveness and sustainable growth. In the conceptual scheme we introduce, socioeconomic organisations can achieve their overall RASI goals through three-level syntheses:

A. Crisis–Innovation–Change Management. The organisation must understand a) that crises are evolutionarily inevitable and irreversible, b) that innovations are providing solutions to old or new problems to exit from crises, and c) that effective change management is a prerequisite

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for innovation but also for the continuous assimilation of the transformations that these innovations bring (see also Chatzinikolaou & Vlados, 2022a).

B. Strategy–Technology–Management. The organisation must understand its core philosophies and processes —in terms of Stra.Tech.Man organisational physiology— enhancing its competitive strengths and innovation potential (see also Vlados & Chatzinikolaou, 2019a, 2020).

C. Human Resource Management (HRM)–Education and Training–Innovation. The organisation must achieve the organic synthesis of HRM with the Stra.Tech.Man core by emphasising continuous in-business and extra-business training of stakeholders to innovate (see also Vlados & Chatzinikolaou, 2020).

3. METHODS

This study integrates one qualitative and three mixed surveys in Eastern Macedonia–Thrace. The inductive and empirically grounded research in these studies aimed to interpret specific behaviours of individuals in this area by interviewing them in their business setting and creating interactional conditions (Corbin & Strauss, 2015; Neergaard & Ulhøi, 2007). We followed a variant of Lindlof and Taylor's (2002) qualitative analysis, designing different interview schedules, sometimes accompanied by psychometric scales and open-ended questions, turning the survey into a mixed-methods approach. According to Lindlof and Taylor (2002, p. 195), the interview schedule "is one way to increase the reliability and credibility of data." Table 1 shows the number of firms in the sample, divided according to the number of employees.

<Table 1 about here>

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Table 1. The sample

Empirical scope	Type	1-10	11-50	20-250	50-250	>251	Sector
Crisis–Innovation–Change Management	Qualitative	38	17				Services
Strategy–Technology–Management	Mixed			45*		4	Mainly services
Strategy Perception and Implementation	Mixed	44**	10				Services
HRM–Education and Training–Innovation	Mixed	45	25		2		Retail

*Most of them employ 20-30 people.
**Primarily 1 to 5 workers.

The sample consists of 230 cases and includes mainly micro and small enterprises (1–10 and 11-50 employees) in the services sector. We picked primarily less competitive companies through our knowledge from years of living and research in this area —conventional sampling that turned purposive subsequently (McCrae & Purssell, 2016). Like Yitshaki and Kropp (2016), we assigned undergraduates of the local university to meet with the managers or owners of the firms, ask them specific questions and sound record all answers. We also instructed the interviewers to ask for further reactions when the respondent's understanding seemed insufficient to cover the "why" behind each question (a laddering-type approach, Corbridge et al., 1994). We designed these *in vivo* studies according to the qualitative research principles of Corbin and Strauss (2015), who contend that emergence and interaction are more important than what is already known.

The four sets of interviews were conducted between 2017 and 2019. The "Crisis–Innovation–Change Management" triangle was the research base that posed the questions related to these respective dimensions, as Figure 1 shows (see also Chatzinikolaou & Vlados, 2022a). The "Strategy–Technology–Management" physiology questionnaire explored extended forms of the Stra.Tech.Man core (see also Vlados & Chatzinikolaou, 2019a). The empirical study concerning the physiological exploration of strategy examined dimensions of perception, planning, and implementation —in greater depth than the Stra.Tech.Man framework and for one of its three aspects (see also Vlados & Chatzinikolaou, 2019b). This examination of strategic perception and implementation contrasted fundamental contributions to the literature (e.g., Hamel & Prahalad, 1989; Mintzberg, 1987; Peters & Waterman, 1982;

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Porter, 1980, 1990). Finally, the triangle "HRM–Education and Training–Innovation" was the theoretical background for the empirical examination of problems concerning the companies' human resources and why their evolution is linked to organisational physiology (see also Vlados & Chatzinikolaou, 2020).

All surveys aimed at comparing —in the form of self-evaluation, as a variant of action research (e.g., Ross, 1998; Winter, 1982)— the past situation with the current perception of a given aspect or problem. Section 3 first presents relevant macro, meso, and micro evidence on the Region of Eastern Macedonia–Thrace (REMTh) and then analyses the findings of these studies in terms of organisational physiology. Section 4 interprets these results, approaching the issue from a critical perspective and based on "thick description" principles that can illuminate causes underlying qualitative research results. Ryle (2009) introduced this approach to distinguish the difference between observations of surface behaviours (thin) and contextual conditions that lead to the occurrence of these interactions (thick). According to Ponterotto (2006), a researcher uses this approach to subjectively explain why certain human behaviours of the studied cases occur during data collection. This approach utilises ethnographic and other relevant historical dimensions to interpret the experience. Finally, raw data from these four studies are available upon request in spreadsheets. All information remains confidential, as we promised our respondents during the interviews.

4. RESULTS

4.1. *The business ecosystem of the REMTh*

This area constitutes one of the 13 NUTS 2 regions in Greece, a country with structural competitiveness problems worsened by the decade-long socioeconomic crisis (e.g., Andrikopoulos & Nastopoulos, 2015). Greek firms appear to be less competitive than their Western European counterparts, and these comparative weaknesses have significantly exacerbated the Greek economy's prolonged fiscal and structural crisis (e.g., Giotopoulos et

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al., 2017). According to the latest Global Entrepreneurship Monitor (2022), confidence in new ventures is still low despite the recent high performance of the Greek economy —this fact illustrates competitiveness deficiencies. GDP per capita decreased by 22% in the REMTh between 2010 and 2019, while in the Greek economy by 12% (see www.statistics.gr). This unfavourable development mainly occurred because the REMTh is a border region whose productive fabric is based primarily on agricultural activities. Figure 2 illustrates this comparative lag of the REMTh, showing its position among the NUTS 2 areas of the EU.

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<Figure 2 about here>

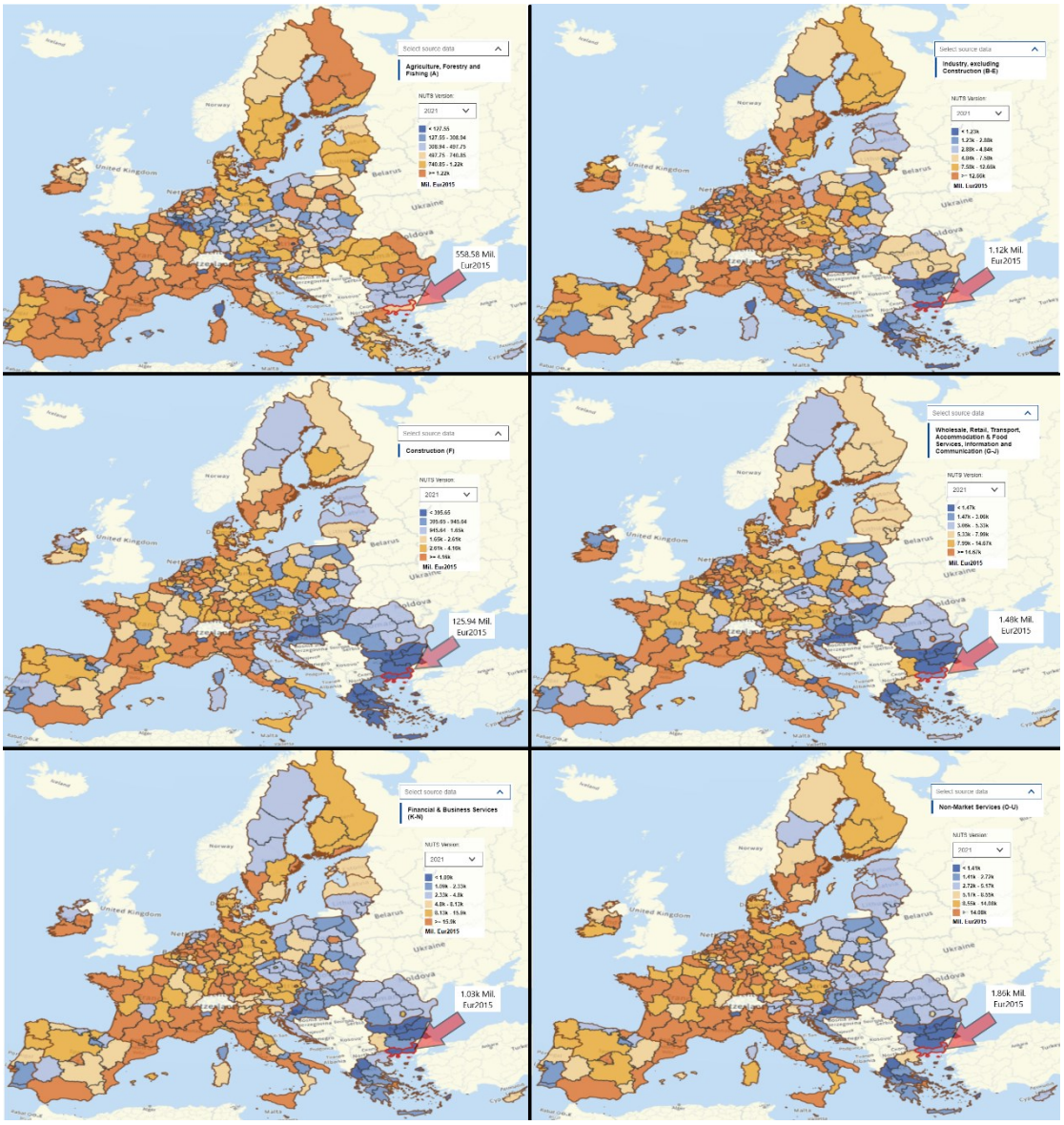


Figure 2. Gross value added of NACE 2 sectors (comparison with EU, constant 2015 prices in EUR). Own elaboration, based on the following: EU Urban Data Platform Plus, https://urban.jrc.ec.europa.eu

The REMTh performs above average only in Agriculture, Forestry, and Fishing. The private services sector (G–J, K–N) —which mainly concerns the sample considered in the next section— performs poorly. Gross value added is also at the lowest relative levels in manufacturing and construction. Simultaneously, non-private services perform slightly better due to relatively larger budgets in defence and education (cf. Boden, 2015). The Agriculture,

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Forestry and Fishing sector is proportionally more specialised than the other industries even though it records a lower amount of gross value added.

This discrepancy is the main structural problem of competitiveness in the REMTh, leading to symptoms of less dynamic sustainable growth as local entrepreneurship is often done by traditional methods rather than systematically. According to the official development strategy of the Managing Authority (Region of Eastern Macedonia–Thrace, 2014), the agricultural sector, which is the main field of specialisation in the region, has low competitiveness due to a lack of modern entrepreneurship methods. The same report notes that the manufacturing sector has similar structural problems due to most companies' relatively small size and family character (see monad-centredness). Also, tourist firms, which constitute a significant part of regional GDP, face chronic comparative problems of introversion and lack of dynamism (Region of Eastern Macedonia–Thrace, 2014).

As a result, innovation performance in the REMTh is among the lowest compared to all other European regions. According to the most recent Regional Competitiveness Index (Annoni & Dijkstra, 2019), the area ranks 265th among the 268 EU28 and 12th among the 13 Greek regions. According to the most recent Regional Innovation Scoreboard (2021), the REMTh belongs to the "emerging innovators," with a relatively strong performance in some innovative business processes and extremely weak in the employment of ICT professionals. According to an in-depth study on the relationship between smart-specialisation structures and higher education in the REMTh (European Commission: Joint Research Centre, 2021), the local economy is over-dependent on low-productivity and low-technology business functions. These deficiencies in organisational development seem to prevent proactive links with the local university —which provides education in a wide range of disciplines.

Boden (2017) presents in-depth guidelines for regional specialisation in the REMTh, which emerged through field surveys and focus-group sessions with stakeholders (Foray, 2014). The

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author suggests that the Managing Authority should invest more vigorously in helping the internationalisation of local firms, focusing on specific market outlets. Boden (2017) also urges regional quadruple-helix actors in the REMTh to formulate strategies based on value chains from existing innovation infrastructures and capabilities instead of focusing on potential comparative disadvantages of the region.

4.2. *The physiological evolution of less competitive firms in the REMTh*

Therefore, based on the business morphology of the REMTh, we expect to find robust evidence of monad-centredness in most firms operating in the secondary or tertiary sector. In the following analysis, we present findings of relevant field studies that converge toward this direction. Below, we list the results of questionnaires on how small entrepreneurs perceive different composite crisis, innovation, and change management dimensions, focusing on conclusions emerging from these findings (see also Chatzinikolaou and Vlados 2022a).

I) Crisis: For these small entrepreneurs, a structural upheaval seems to mean —almost entirely— something that arises mainly as an external conjunctural change that causes reduced revenues and demand due to increased taxation, excessive borrowing, and intensified competition. In the perception of these individuals, the national government seems to be mainly responsible and should reduce tax rates, increase subsidies, and strengthen social welfare mechanisms.

II) Innovation: When asked about dimensions of innovative progress, these small entrepreneurs perceive an extraordinary improvement —mainly because of luck and inspiration. They argue that this organisational improvement does not seem directly related to their day-to-day operations. To a significant extent, these individuals do not appear competent in describing how they have innovated recently —they consider their low innovativeness almost exclusively due to unfavourable conditions in the external business environment. In the perception of these small entrepreneurs, ineffective government policies are the perpetuating

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cause of these obstacles, proposing as a solution the facilitation by banks of loans with lower lending rates and the further diffusion of European business support programmes.

III) Change management: The prevailing perception is that “everything remains unchanged in the background” for these small entrepreneurs. They perceive change management —for the most part— as an immediate and short-term response to external adverse fluctuations. These individuals show that they adapt mainly by demonstrating remarkable resilience —they seem willing to limit their ambitions and the firm’s profitability to survive. They also exhibit a relative lack of understanding and implementation of systematic change management processes.

These small entrepreneurs in the service sector of the REMTh seem to perceive the crises as mainly quasi-external phenomena that the national government can cure with sectoral protection, lower taxation, and enhanced aggregate demand. In terms of innovation, these organisations appear to have distinct shortcomings leading to lower profitability and competitiveness. They also do not seem to implement —as much and systematically— change management methodologies, which are imperative for overcoming crises.

We present below identified inefficiencies of small entrepreneurs in different compositional dimensions of strategy, technology, and management (see also Vlados & Chatzinikolaou, 2019a).

A) Strategy: These less competitive organisations mostly exhibit limited vertical integration dynamics and negligible sectoral diversification. Also, the firm’s geographic expansion is mainly local, leading to difficulties in finding suppliers and financial resources with a broader geographic reach. The information technology integration strategy and the overall production process are also mainly local. Finally, these small entrepreneurs appear to a large extent not to develop the human resources of their enterprises strategically.

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B) Technology: We find a relative lack of systematisation due to comparative deficiencies in the overall technological perspective of the organisation. We also see comparatively limited investment in facilities and general modernisation. These small entrepreneurs seem to use new technology to a relatively small extent to improve productivity, resulting in neither creating nor exploiting niches of high-value-added knowledge. They also show limited technological superiority over competitors, as the range of specialised products is usually small. The product core also appears to lack specialisation, a trend reflected in information technology's comparatively diminished use and development.

C) Management: These small entrepreneurs seem to face comparative difficulties in developing a modern management philosophy —systematisation is limited, mainly due to the family nature of the enterprises. Thus, we find relatively little investment in managerial skills and limited use of modern and advanced forms of leadership. These comparative shortcomings are also evident from the relative absence of formal human resource development processes, corporate social responsibility mechanisms, and total quality management approaches. Overall, outward entrepreneurial orientation through managerial innovations in these organisations appears limited.

By distributing questionnaires with psychometric scales to these small entrepreneurs, we saw that they rated these different "Strategy–Technology–Management" dimensions higher in the present than in the past —the paradox is that we identified comparative inadequacies from our in-depth qualitative review of the cases. Overall, we found an effort of strategic refocusing, technological modernisation, and systematisation of management, which does not seem to be successful for the required innovative transition in organisational physiology and structure.

We present below how small entrepreneurs perceive and implement more specific strategic concepts compared to the strategy-technology-management synthesising approach we field-tested in the previous survey (see also Vlados & Chatzinikolaou, 2019b).

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(i). These companies show relative inadequacy in establishing and monitoring a principal strategic goal.

(ii). Their general conception of strategy does not appear to be much oriented toward flexibility in dealing with change.

(iii). We find an almost universal absence of a written mission statement.

(iv). To a large extent, the strategy does not consider all the stakeholders in structuring the vision and mission.

(v). Most of these organisations do not seem to exploit extra-sectoral trends to make strategic choices —see PEST synthesis (exploring political, economic, social, and technological factors).

(vi). These firms do not monitor —as much and systematically— the external environment using Porter’s (1980) five forces analysis.

(vii). These firms do not seem to reinforce —to a significant extent— their competitiveness through strategic interaction with the local external environment.

(viii). They do not appear to be doing much strategic benchmarking by comparing successful practices within and outside their field of activity.

(ix). They do not appear to typically implement an internal audit strategy through systematic monitoring and evaluation of comparative strengths.

(x). These firms do not extensively use Mintzberg’s (1987) 5P approach to strategy (plan, ploy, pattern, position, or perspective).

(xi). To a large extent, they do not seem to proactively build core competencies and strategic intent by changing the game rules.

(xii). Their strategic elaboration seems, in many cases, insufficient in creating a synthesising evolving mission–vision–philosophy triangle.

(xiii). Strategic planning does not extensively appear to create comparative advantages in the firm’s day-to-day practice.

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(xiv). Strategic decisions primarily involve the owner without group consultation.

(xv). The generic strategy seems to be based mainly on the saleability of a product rather than on broader social dimensions.

(xvi). These firms show relative inadequacy in diffusing organisational excellence practices throughout business processes (cf. Peters & Waterman, 1982).

By distributing questionnaires with psychometric scales to these small entrepreneurs, we saw that they rated these specific strategic dimensions higher in the present compared to the past —the paradox is that we identified comparative inadequacies from our in-depth qualitative review of the cases. As in the field research on the Strategy–Technology–Management triangle, these small entrepreneurs claimed to be trending towards a more systematic implementation of their business strategy —yet they did not seem to justify this rapid progress adequately. By examining 16 critical strategic questions, these firms seemingly have not recently innovatively transformed their organisational physiology.

Regarding the last step of these complementary surveys on the physiological characteristics of specific organisations, we present below the results of questionnaires on how small entrepreneurs perceive different compositional dimensions of HRM, education–training, and innovation (see also Vlados & Chatzinikolaou, 2020).

(a) Human Resource Management: These individuals have relatively ambiguous perceptions of the personal characteristics of employees. On the one hand, they seem interested in staff experience and training. On the other hand, they claim, to some extent, that nothing has changed in how they manage their people and that they do not apply specific relevant selection criteria. They also show ambiguous perceptions —to a significant degree— of the difficulties they face in HRM. On the one hand, they are concerned about high salary costs, insufficient staff expertise, and work inexperience. On the other hand, they largely fail to clarify why these

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problems are essential for their businesses, stating —in a quasi-static way— that they require minimal skills from their staff.

(b) Education and Training: Equally fluctuating are their perceptions of education and training processes. On the one hand, they show eagerness to employ people with high skills and relevant work experience. On the other hand, they appear to underutilise their staff’s higher education and training, requiring them to perform low-skilled tasks. Thus, although they seem to be sensitive toward the personal development of their employees, they do not seem to take much initiative —formal or informal.

(c) Innovation: The perceptions of these small entrepreneurs are also, to some extent, contradictory on innovation issues. For example, most small entrepreneurs say that their value on innovation increases over time and that there is —in essence— no innovative progress in small retail firms like theirs. Therefore, to a significant extent, we find a relative absence of a link between innovation, HRM, and education–training in the perceptions of these small entrepreneurs.

By distributing questionnaires with psychometric scales to these individuals, we saw that they rated all the compositional dimensions of HRM, education–training, and innovation higher in the present compared to the past —the paradox is that we identified comparative inadequacies from the in-depth qualitative review of the cases. As in the previous surveys, these small entrepreneurs claimed that they have recently tended toward the systematization of their HRM, education–training, and innovation processes. However, the findings point to a different picture.

In the previous section, we showed that innovation performance in the REMTh—particularly concerning service sector firms—is among the lowest compared to the EU. Our findings seemingly confirm that the relative dominance of monad-centredness in firms can lead the entire business ecosystem to comparative underdevelopment (see competitiveness deficits).

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We also see yet another paradox. The physiological transformation of these firms —in recent years, according to the self-assessments we studied— shows that it simultaneously involves elements of massification and monad-centredness. On the one hand, these firms appear to desire a transformation mainly toward increased massiveness. On the other hand, they remain relatively captive to their existing competitiveness potential. The findings of these field studies revealed discrepancies regarding this desire, and the next section discusses in depth why this is the case.

5. CRITICAL DISCUSSION AND CONCLUSIONS

In this study, we presented the results of different questionnaires that preliminarily diagnosed the organisational physiology of specific firms. After this synthesis, it appears that the conclusions of each survey are enriched as the findings complement each other. We saw that most small entrepreneurs in a less developed region exhibited a relatively poor perception of the Crisis–Innovation–Change Management triangle compared to contemporary business development theory. Regarding the Stra.Tech.Man physiology of these firms, we also found a gap in the perception of these small entrepreneurs compared to organisational development theory. Most of them also seemed to perceive the triangle of HRM–Education and Training–Innovation in a somewhat fragmented way, as they did not seem able to elaborate in depth on the conceptual and practical prerequisites of this evolutionary link. Therefore, we conclude that the organisational physiology approach enables us to delve into problems of competitiveness and sustainability in less developed business ecosystems, helping to analyse dimensions of resilience in the new globalisation that go deeper than the mere size of the firm. Improving an entrepreneurial ecosystem seems to depend on the physiological strengthening of locally established firms. In the following section, we delve into this insight.

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5.1. *The organisational physiology as a diagnostic approach and policy implications*

These preliminary physiological diagnoses are an "x-ray" of the REMTh, highlighting specific problems of organisational underdevelopment and future evolutionary prospects. Based on a thick description, we can examine some of our qualitative findings from a different —more profound— perspective. During data collection, the respondents seemed to embellish their answers to some extent, fearing possible negative criticism from the local community, which is the primary clientele. Many small entrepreneurs also gave limited in scope and depth responses when the interviewers sometimes asked for clarification during the interview. One explanation for this behaviour is based on the relative lack of knowledge that the small entrepreneurs had about the issues under consideration. We also note again that we had an earlier heightened sense of the competitiveness problems that the REMTh seems to be facing, which allowed us to quickly understand which firms were less and more competitive. In support of this hypothesis, several of these less-dynamic firms under consideration appear to be inactive today. Based on the initial objectives, we conclude the following:

First, we see a desire for systematisation with solid evidence of monad-centredness in the physiological transformation of these firms. This contradiction starts from the monad-centric physiological core (in Stra.Tech.Man terms). The perceptions of these small entrepreneurs seem to confirm that monad-centric firms implement their strategy mainly on instinct, often make technological decisions sporadically, and manage organisational resources primarily based on their day-to-day experience. *Prima facie*, we find a relative absence of fundamental business organisation knowledge in these small entrepreneurs. Another explanation is their relatively limited growth ambitions —to some extent, they seem to disbelieve that entrepreneurship means the aspiration to create a "private kingdom" (cf. Schumpeter, 1934). Concerning their physiological transformation, another finding during data collection is that most respondents showed a relative "teacher-like disposition" toward the interviewers (who were mostly

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younger). One explanation for this behaviour is the introversion that characterises these firms, which led these small entrepreneurs to seek relative concealment of their weaknesses, trying to "go on the offensive" as they felt criticised. Therefore, we conclude that these less competitive firms transformed for the worse —in terms of physiology— during the Greek crisis[i].

Second, the research that first introduced the Stra.Tech.Man physiology seems to be reaffirmed (see Vlados, 2004), as it argued that most Greek firms are monad-centred, leading the entire socioeconomic system to a problematic adaptation to globalisation. These four field surveys we conducted in the REMTh show an apparent dominance of monad-centric entrepreneurship forms, especially in low-value-added sectors (cf. Figure 2). They also reflect the challenges of adapting to the new globalisation; the achievement of RASI objectives seems increasingly distant for these enterprises, making the REMTh —*lato sensu*— relatively weak regarding its sustainable development trajectory.

Third, as it seems, the small entrepreneurs of such weak firms (in organisational dynamics) tend to invent external enemies to cover up their competitive shortcomings. They do not seem to analyse in depth their internal weaknesses (due to introversion), exhausting themselves on the external threats that are reasonably multiplying for them. Hence, traditional —mainly neoclassical or neo-Keynesian-oriented— macro–meso policies for the horizontal reinforcement of spatial entities are becoming increasingly ineffective nowadays (cf. Chatzinikolaou et al., 2021; Nelson, 2018). Since physiological differences are subtle and complex, approaches that see homogenous firms in specific sectors appear increasingly myopic (cf. Gomes & Gubareva, 2021). Ostensibly, entrepreneurship support policies today should focus on providing business development tools —not simply stimulating demand with cheap or "free money"— to help these organisations hybridise into more flexible forms of Stra.Tech.Man physiology.

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If there are enough sick people, the state must ensure that hospitals are built. Similarly, in economies with many less competitive firms, policymakers should ensure that they provide mechanisms to strengthen their physiology —i.e., resources for hiring professional consultants or providing in-house advice (e.g., Adam-Smith & McGeever, 1995). We contend that the Institutes for Local Development and Innovation —a mechanism for intervention and policy articulation introduced by Chatzinikolaou and Vlados (2021) but not yet implemented on the ground— is a step in the right direction. As Chatzinikolaou and Vlados (2021) argue, the aim of this proposed entrepreneurship support institution in regional entrepreneurial ecosystems could be to interconnect actors that can foster entrepreneurship and that are uncoordinated in a region, providing "Stra.Tech.Man business clinic" services. This institution could be a mechanism to strengthen less competitive organisations in the REMTh, as the diagnosis of physiology presented in this paper could be a form of service provision. Such an intervention at the meso–micro level could monitor its development environment and take initiatives to coordinate relevant actors. It could also offer free advice through grants or own interventions (essentially being a "business clinic") and publish region-wide development results periodically (e.g., annually).

5.2. *Limitations and prospects*

Inevitably, the present paper is subject to some limitations that must be mentioned but are nonetheless potential prospects for future research. The physiological transformation we have studied concerns developments that preceded the COVID-19 pandemic and the Russian invasion that is currently changing the global energy and geo-economic map. Therefore, we believe an update of the research could shed light on new aspects. It also did not consider the apparent high performance of the Greek economy after its crisis and the increasingly important role that the REMTh is nowadays acquiring due to the strategic importance of the region's ports for the Western alliance and the US. Also, the findings are inevitably limited to the REMTh.

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Future research could explore the physiological transformation of other areas —possibly more (or less) developed European–Greek regions. Such a re-approach would probably make the findings more reliable. Finally, a form of pilot support for some enterprises in the form of free consultancy could serve as a springboard to see whether a policy for supporting them with such tools would make sense as it would strengthen their physiology. However, most of the less capable entrepreneurs —in terms of integrating their businesses into a sustainable growth spiral of the region— said that they do not wish to cooperate further with our team, which makes their empowerment an intractable problem.

REFERENCES

Adam-Smith, D., & McGeever, M. (1995). Training and Enterprise Councils and small- and medium-sized enterprises: Not meeting needs or just not meeting the customer? *Journal of Small Business and Enterprise Development, 2*(3), 149–157. https://doi.org/10.1108/eb020957

Andrikopoulos, A., & Nastopoulos, Ch. (2015). *Κρίση και ρεαλισμός [Crisis and Realism]*. Propobos Publications.

Annoni, P., & Dijkstra, L. (2019). *The EU Regional Competitiveness Index 2019*. Publications Office of the European Union.

Belyaeva, Z., & Lopatkova, Y. (2020). The impact of digitalization and Sustainable Development Goals in SMEs' strategy: A multi-country European study. In A. Thrassou, D. Vrontis, Y. Weber, S. M. R. Shams, & E. Tsoukatos (Eds.), *The Changing Role of SMEs in Global Business: Volume II: Contextual Evolution Across Markets, Disciplines and Sectors* (pp. 15–38). Springer International Publishing. https://doi.org/10.1007/978-3-030-45835-5_2

Bhattacharya, A., Khanna, D., Schweizer, C., & Bijapurkar, A. (2017, April 25). The new globalization: Going beyond the rhetoric. *BCG Henderson Institute*.

This is the Accepted Manuscript (AM, also known as "post-print").This extract is taken from the author's original manuscript and has not been reviewed or edited. The definitive published version of this extract may be found at https://link.springer.com/10.1007/978-3-031-37361-9_10.

https://www.bcg.com/en-gr/publications/2017/new-globalization-going-beyond-rhetoric.aspx

Boden, M. (2017). RIS3 Implementation in Lagging Regions: Lessons from Eastern Macedonia and Thrace. *European Structural & Investment Funds Journal, 5*(1), 77–83.

Boden, M., dos Santos, P., Haegeman, K., Marinelli, E., & Valero, S. (2015). *European Parliament Preparatory Action: Actual and desired state of the economic potential in regions outside the Greek capital Athens* (EUR 27570). Publications Office of the European Union. https://doi.org/10.2791/34670

Chatzinikolaou, D., & Vlados, Ch. (2019). Evolutionary economics and the Stra.Tech.Man approach of the firm into globalization dynamics. *Business, Management and Economics Research*, *5*(10), 146–160. https://doi.org/10.32861/bmer.510.146.160

Chatzinikolaou, D., & Vlados, Ch. (2022a). Crisis, innovation and change management: A blind spot for micro-firms? *Journal of Entrepreneurship in Emerging Economies*. (forthcoming)

Chatzinikolaou, D., & Vlados, Ch. (2022b). The world system's mutational crisis and the emergence of the new globalization. In B. Lamotte (Ed.), *Économie sociale et crises du XXIe siècle* (pp. 163–192). L'Harmattan.

Chatzinikolaou, D., Demertzis, M., & Vlados, Ch. (2021). European entrepreneurship reinforcement policies in macro, meso, and micro terms for the post-COVID-19 era. *Review of European Studies, 13*(2), 39–56. https://doi.org/10.5539/res.v13n2p39

Corbin, J. M., & Strauss, A. L. (2015). *Basics of qualitative research: Techniques and procedures for developing grounded theory*. Sage.

Corbridge, C., Rugg, G., Major, N. P., Shadbolt, N. R., & Burton, A. M. (1994). Laddering: Technique and tool use in knowledge acquisition. *Knowledge Acquisition*, *6*(3), 315–341. https://doi.org/10.1006/knac.1994.1016

This is the Accepted Manuscript (AM, also known as "post-print").This extract is taken from the author's original manuscript and has not been reviewed or edited. The definitive published version of this extract may be found at https://link.springer.com/10.1007/978-3-031-37361-9_10.

Dopfer, K., Foster, J., & Potts, J. (2004). Micro-meso-macro. *Journal of Evolutionary Economics*, *14*(3), 263–279. https://doi.org/10.1007/s00191-004-0193-0

European Commission: Joint Research Centre. (2021). *Higher Education for Smart Specialisation: The Case of Eastern Macedonia and Thrace*. Publications Office. https://data.europa.eu/doi/10.2760/612671

Foray, D. (2014). From smart specialisation to smart specialisation policy. *European Journal of Innovation Management*, *17*(4), 492–507. https://doi.org/10.1108/EJIM-09-2014-0096

Foster, J. (1997). The analytical foundations of evolutionary economics: From biological analogy to economic self-organization. *Structural Change and Economic Dynamics, 8*(4), 427–451. https://doi.org/10.1016/S0954-349X(97)00002-7

Giotopoulos, I., Kontolaimou, A., & Tsakanikas, A. (2017). Antecedents of growth-oriented entrepreneurship before and during the Greek economic crisis. *Journal of Small Business and Enterprise Development*, *24*(3), 528–544. https://doi.org/10.1108/JSBED-01-2017-0003

Global Entrepreneurship Monitor. (2022). *Global Entrepreneurship Monitor 2021/2022 Global Report: Opportunity Amid Disruption*. GEM.

Gomes, O., & Gubareva, M. (2021). Complex Systems in Economics and Where to Find Them. *Journal of Systems Science and Complexity, 34*(1), 314–338. https://doi.org/10.1007/s11424-020-9149-1

Hamel, G., & Prahalad, C. K. (1989). Strategic intent. *Harvard Business Review, 67*(3), 63–76.

Hodgson, G., & Knudsen, T. (2004). The firm as an interactor: Firms as vehicles for habits and routines. *Journal of Evolutionary Economics, 14*(3), 281–307. https://doi.org/10.1007/s00191-004-0192-1

Karagouni, G. (2018). The role of Dynamic Entrepreneurial Capabilities and innovation in intergenerational succession of family firms. In D. Vrontis, Y. Weber, A. Thrassou, S. M.

This is the Accepted Manuscript (AM, also known as "post-print").This extract is taken from the author's original manuscript and has not been reviewed or edited. The definitive published version of this extract may be found at https://link.springer.com/10.1007/978-3-031-37361-9_10.

R. Shams, & E. Tsoukatos (Eds.), *Innovation and Capacity Building: Cross-disciplinary Management Theories for Practical Applications* (pp. 31–54). Springer International Publishing. https://doi.org/10.1007/978-3-319-90945-5_3

Killian, L. J. (2021). Taming the dark side of the new globalization. In S. H. Park, M. A. Gonzalez-Perez, & D. E. Floriani (Eds.), *The Palgrave Handbook of Corporate Sustainability in the Digital Era* (pp. 355–376). Springer International Publishing. https://doi.org/10.1007/978-3-030-42412-1_18

Lampadarios, E., & Kyriakidou, N. (2020). Critical Success Factors for SMEs in the UK chemical distribution industry: The 'Brexit' Effect. In A. Thrassou, D. Vrontis, Y. Weber, S. M. R. Shams, & E. Tsoukatos (Eds.), *The Changing Role of SMEs in Global Business: Volume I: Paradigms of Opportunities and Challenges* (pp. 155–177). Springer International Publishing. https://doi.org/10.1007/978-3-030-45831-7_8

Lindlof, T. R., & Taylor, B. C. (2002). *Qualitative communication research methods* (2nd ed.). Sage Publications.

Marshall, A. (1890). *Principles of economics*. Macmillan.

McCrae, N., & Purssell, E. (2016). Is it really theoretical? A review of sampling in grounded theory studies in nursing journals. *Journal of Advanced Nursing*, *72*(10), 2284–2293. https://doi.org/10.1111/jan.12986

Migliaccio, G., & Rossetti, L. U. (2020). Italian Furniture Sector SMEs: Sustainability and Commercial Ethics. In A. Thrassou, D. Vrontis, Y. Weber, S. M. R. Shams, & E. Tsoukatos (Eds.), *The Changing Role of SMEs in Global Business: Volume II: Contextual Evolution Across Markets, Disciplines and Sectors* (pp. 225–259). Springer International Publishing. https://doi.org/10.1007/978-3-030-45835-5_11

Mintzberg, H. (1987). The strategy concept I: Five Ps for strategy. *California Management Review, 30*(1), 11–24. https://doi.org/10.2307/41165263

This is the Accepted Manuscript (AM, also known as "post-print").This extract is taken from the author's original manuscript and has not been reviewed or edited. The definitive published version of this extract may be found at https://link.springer.com/10.1007/978-3-031-37361-9_10.

Montanari, S., & Kocollari, U. (2020). Defining the SME: A Multi-Perspective Investigation. In A. Thrassou, D. Vrontis, Y. Weber, S. M. R. Shams, & E. Tsoukatos (Eds.), *The Changing Role of SMEs in Global Business: Volume II: Contextual Evolution Across Markets, Disciplines and Sectors* (pp. 61–82). Springer International Publishing. https://doi.org/10.1007/978-3-030-45835-5_4

Moore, J. (1993). Predators and prey: A new ecology of competition. *Harvard Business Review*, *71*(3), 75–86.

Neergaard, H., & Ulhøi, J. P. (Eds.). (2007). *Handbook of qualitative research methods in entrepreneurship*. Edward Elgar.

Nelson, R., & Winter, S. (1982). *An evolutionary theory of economic change*. The Belknap Press of Harvard University Press.

Nelson, R., Dosi, G., Helfat, C., Winter, S., Pyka, A., Saviotti, P., Lee, K., Malerba, F., & Dopfer, K. (2018). *Modern evolutionary economics: An overview*. Cambridge University Press.

Onyama, D. N. (2021). *Profitability, productivity, and sustainability: Organizational behavior and strategic alignment*. Routledge.

Peters, T. J., & Waterman, R. H. (1982). *In search of excellence: Lessons from America's best-run companies*. Harper & Row.

Ponterotto, J. (2006). Brief note on the origins, evolution, and meaning of the qualitative research concept thick description. *The Qualitative Report*, *11*(3), 538–549. https://doi.org/10.46743/2160-3715/2006.1666

Porter, M. (1980). *Competitive strategy: Techniques for analyzing industries and competitors*. Free Press.

Porter, M. (1990). *The competitive advantage of nations*. Free Press.

This is the Accepted Manuscript (AM, also known as "post-print").This extract is taken from the author's original manuscript and has not been reviewed or edited. The definitive published version of this extract may be found at https://link.springer.com/10.1007/978-3-031-37361-9_10.

Region of Eastern Macedonia–Thrace. (2014). *Το Επιχειρησιακό Πρόγραμμα «Ανατολική Μακεδονία και Θράκη» 2014 – 2020 [The Operational Programme "Eastern Macedonia and Thrace" 2014–2020]*. https://www.eydamth.gr/index.php/extras/to-programma

Regional Innovation Scoreboard. (2021). *Regional innovation scoreboard 2021: Regional profile of Greece*. https://research-and-innovation.ec.europa.eu/statistics/performance-indicators/regional-innovation-scoreboard_en

Rinkinen, S., & Harmaakorpi, V. (2018). The business ecosystem concept in innovation policy context: Building a conceptual framework. *Innovation: The European Journal of Social Science Research*, *31*(3), 333–349. https://doi.org/10.1080/13511610.2017.1300089

Ross, J. A., Rolheiser, C., & Hogaboam-Gray, A. (1998). Skills training versus action research in-service: Impact on student attitudes to self-evaluation. *Teaching and Teacher Education*, *14*(5), 463–477. https://doi.org/10.1016/S0742-051X(97)00054-1

Ryle, G. (2009). The thinking of thoughts: What is 'Le Penseur'doing? In *Collected Essays 1929-1968* (pp. 516–532). Routledge.

Schumpeter, J. (1934). *The theory of economic development: An inquiry into profits, capital, credit, interest, and the business cycle* (R. Opie, Trans.; Reprint Edition: New Brunswick, N.J, Transaction Books, 1983, Social science classics series). Harvard University Press.

SEVE. (2018). *Χαρτογράφηση της εξαγωγικής δραστηριότητας της Ελλάδας ανά περιφέρεια [Mapping the export activity of Greece per region]*. Greek Exporters Association. https://www.seve.gr/wp-content/uploads/2019/12/ΠΕΡΙΦΕΡΕΙΑΚΑ_2014-2018_Short-Version.pdf

Teece, D. J., Pisano, G., & Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal, 18*(7), 509–533. https://doi.org/10.1002/(SICI)1097-0266(199708)18:7<509::AID-SMJ882>3.0.CO;2-Z

This is the Accepted Manuscript (AM, also known as "post-print").This extract is taken from the author's original manuscript and has not been reviewed or edited. The definitive published version of this extract may be found at https://link.springer.com/10.1007/978-3-031-37361-9_10.

Vargas-Hernández, J. G., & Orozco Quijano, E. P. (2022). Business under crisis: Strategic organizational sustainability— a contextual transformation. In D. Vrontis, A. Thrassou, Y. Weber, S. M. R. Shams, E. Tsoukatos, & L. Efthymiou (Eds.), *Business Under Crisis, Volume III: Avenues for Innovation, Entrepreneurship and Sustainability* (pp. 19–45). Springer International Publishing. https://doi.org/10.1007/978-3-030-76583-5_2

Vlados, Ch. (2004). *La dynamique du triangle stratégie, technologie et management: L'insertion des entreprises grecques dans la globalisation [The dynamics of the triangle of strategy, technology and management: The insertion of Greek enterprises into globalisation]* [Thèse de doctorat de Sciences Économiques, Université de Paris X-Nanterre]. http://www.theses.fr/2004PA100022

Vlados, Ch., & Chatzinikolaou, D. (2019a). Business ecosystems policy in Stra.Tech.Man terms: The case of the Eastern Macedonia and Thrace region. *Journal of Entrepreneurship, Management and Innovation, 15*(3), 163–197. https://doi.org/10.7341/20191536

Vlados, Ch., & Chatzinikolaou, D. (2019b). Strategy perception and implementation on less developed business ecosystems micro and small enterprises: The service sector of Eastern Macedonia and Thrace. *Business Management and Strategy, 10*(2), 27–63. https://doi.org/10.5296/bms.v10i2.15008

Vlados, Ch., & Chatzinikolaou, D. (2020). Stra.Tech.Man innovation, HRM and perception of educational needs in underdeveloped business ecosystems: The case of retail sector firms in Eastern Macedonia and Thrace. *International Journal of Human Resource Studies, 10*(2), 330–354. https://doi.org/10.5296/ijhrs.v10i2.17139

Vlados, Ch., Koronis, E., & Chatzinikolaou, D. (2021). Entrepreneurship and crisis in Greece from a neo-Schumpeterian perspective: A suggestion to stimulate the development process at the local level. *Research in World Economy*, *12*(2), 1–16. https://doi.org/10.5430/rwe.v12n2p1

This is the Accepted Manuscript (AM, also known as "post-print").This extract is taken from the author's original manuscript and has not been reviewed or edited. The definitive published version of this extract may be found at https://link.springer.com/10.1007/978-3-031-37361-9_10.

Winter, R. (1982). "Dilemma Analysis": A contribution to methodology for action research. *Cambridge Journal of Education*, *12*(3), 161–174. https://doi.org/10.1080/0305764820120303

Witt, U. (2006). Evolutionary concepts in economics and biology. *Journal of Evolutionary Economics, 16*(5), 473–476. https://doi.org/10.1007/s00191-006-0034-4

World Economic Forum. (2018). *Agile governance: Reimagining policy-making in the Fourth Industrial Revolution* [White Paper]. https://www.weforum.org/whitepapers/agile-governance-reimagining-policy-making-in-the-fourth-industrial-revolution/

Yitshaki, R., & Kropp, F. (2016). Motivations and opportunity recognition of social entrepreneurs. *Journal of Small Business Management*, *54*(2), 546–565. https://doi.org/10.1111/jsbm.12157

[i] Nevertheless, there were some exceptional cases of companies that deserve mention. We merely examined the apparent trends for most firms, focusing on less dynamic micro-businesses in the tertiary sector. However, by distributing the same questionnaires to more dynamic exporting firms (some employing more than 250 employees; cf. SEVE, 2018), we saw the distance between less and more developed firms in the REMTh. We also observed that even massive or flexible firms could have monad-centred elements.